

IT7027C Cloud Infrastructure Technologies

Asynchronous 100% Online

Instructor

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Teaching Assistant

TBD

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Course Description

This course introduces the principles and technologies of modern cloud infrastructure systems that enable and sustain the needs of organizations. Students explore how to design, deploy, and manage scalable, secure, and cost-efficient cloud environments across multiple providers. The course emphasizes virtualization, networking, automation, monitoring, and resilience in cloud ecosystems.

Pre-Reqs

None

Credit Hours and Course Format

3 credit hours; Online.

Course Learning Outcomes

Upon completion of this course, the student will be able to:

- Design and implement cloud infrastructure solutions using best practices for scalability, reliability, and cost optimization.
- Configure and evaluate networking, compute, and storage components across multi-cloud environments.
- Automate cloud resource provisioning and monitoring using Infrastructure as Code and management tools.
- Apply security and compliance controls in cloud operations.
- Assess and compare cloud service models (IaaS, PaaS, SaaS) for organizational needs.

Course Vision

In this course, you'll gain essential knowledge and hands-on skills for designing, managing, and troubleshooting cloud infrastructure.

My teaching approach centers on interactive learning and practical applications. In this course, I'll ensure you engage with the material through labs, group discussions, and real-world case studies. Expect a balance of theory and practice. I believe in fostering collaboration and independent thinking, encouraging you to explore, ask questions, and experiment with different storage setups.

Success in this Course

You have worked hard to be here; you belong here. Congratulations on your achievements, and welcome to the course. I am committed to creating and fostering a positive learning and working environment based on open communication and mutual respect. Your success in this class is important to me. We all learn differently and bring different strengths and needs to the class. If there are aspects of the course you're struggling with, please let me know. Together we'll develop strategies to meet your needs and the requirements of the course.

Course Materials

All course materials are on Canvas.

Textbook

Not required

Required Technology

Please check the School of IT minimum computer requirements 2026-2027 in this [link](#).

Course Policies and Grading

Attendance/Participation Expectations

You can expect to spend between four to six hours each week on project assignments and should log in to view Canvas at a minimum of twice per week. You cannot pass this course without submitting your assignments.

To receive an A in this course, you must do “A” work. Do not ask in the final weeks of the semester how to get an A in the course; do quality work starting in week 1.

Group Work Expectations

Typically, each team has two or three students assigned who work together throughout the semester on different tasks and group discussions.

Generative AI Policy

You are allowed to use the university approved Gen-AI tools like MS Co-Pilot, BearcatGPT. However, you need to include what the AI tool was used for, in each instance. If you fail to acknowledge the use of AI properly in any assignment, you will receive no point for that assignment.

Absences

Unexcused absences are not allowed in this course. If you are going to miss the class, let me know as early as you can. Even though this is an online course, attendance is required and can be measured by page views, unique login days, and time spent on the course.

Communication Policy

There are several ways to reach me as listed on the top of this document. E-mail is the preferred communication platform for questions relating to the class. I will always try my best to respond as soon as possible, but it may take up to 48 hours to receive a response at times. Keep this in mind if you plan to complete assignments at the last minute.

Virtual (via Teams) and in person office hours can be scheduled as necessary. Please e-mail the instructor to schedule a suitable time or use the booking link available at the top.

Announcements will be sent through the Learning Management System. Be sure to have an e-mail that you monitor listed to receive announcements in the LMS or check the announcements section often. Announcements often relate to changes in scheduling or

common issue students may have with assignments. Paying attention to announcements may save you a lot of headaches.

Feedback

Feedback will be provided as a comment in Canvas after the assignment is graded within 2 business days after the due date. If an associated rubric is available, it will be found on the assignment page.

Assignments and Submission Expectations

You are expected to complete work on schedule. Deadlines are part of the real-world environment you are being prepared for. You are not asked to submit assignments on the date it is due. I always try to help my students in unforeseeable situations. The normal submission period of a weekly assignment is 7 days. I also give extra 3 days after the deadline to submit with late penalty. So, students can submit on these 10-11 days. However, if you don't submit during the period given, I cannot accept your submission to ensure fairness among all students.

Late or Missing Assignment Policy

You lose 5% every 12 hours. You lose 5% in the first minute of each 12-hour period. If you are late.

00:01 – 12:00 hours: 5% off.

12:01 – 24:00 hours: 10% off.

24:01 – 36:00 hours: 15% off.

36:01 – 48:00 hours: 20% off.

48:01 – 60:00 hours: 25% off.

60:01 – 72:00 hours: 30% off.

More than 72 hours (3 days): No credit given.

Grading Scale

93.00% and above = A

90.00%-92.99% = A-

87.00%-89.99% = B+

83.00%-86.99% = B

80.00%-82.99% = B-

77.00%-79.99% = C+

73.00%-76.99% = C

70.00%-72.99% = C-

67.00%-69.99% = D+

63.00%-66.99% = D

60.00%-62.99% = D-

less than 60.00% = F

It's highly unethical and annoying to bring to the instructor's attention that you need a high grade for any reason. Doing that is equivalent to asking for a free good grade. It's your responsibility to do all the coursework and necessary study to reach the grade that you want.

The course grades are weighted as follows:

Assignments	Points Assigned	Weight
Participation and attendance	10	5%
Weekly discussion	60	25%
Assignments	120	30%
Final project and presentation	200	40%
Extra credit for course evaluation	10	1%

Course Schedule

Module	Dates	Modules/Topics	Due Dates
1	08/24 – 08/30	Cloud Paradigm and Virtualization Foundation	Aug 30
2	08/31 – 09/06	Cloud service and the shared responsibility model	Sep 06
3	09/07 – 09/13	Infrastructure Emulation and API Endpoints	Sep 13
4	09/14 – 09/20	Software-Defined Networking (SDN) Concepts	Sep 20
5	09/21 – 09/27	Scalable Computer and Object Storage	Sep 27
6	09/28 – 10/04	Cloud native data and serverless architecture	Oct 04
7	10/05 – 10/11	Secret Management and Zero Trust	Oct 11
8	10/12 – 10/18	Advanced Infrastructure as Code (IaC)	Oct 18
9	10/19 – 10/25	Containerization and Kubernetes Integration	Oct 25
10	10/26 – 11/01	High Availability and Load Balancing	Nov 01
11	11/02 – 11/08	Multi-cloud environments	Nov 08
12	11/09 – 11/15	Observability and Monitoring	Nov 15
13	11/16 – 11/22	Infrastructure deployment CI/CD pipeline in Github	Nov 22
14	11/23 – 11/29	Resilience in Cloud Systems	Nov 29
15	11/30 – 12/06	Final project and presentation submission	Dec 06

UC Policies

Attendance: Students at the University of Cincinnati are expected to attend classes in order to meet the learning objectives for the course. Students are expected to follow the class attendance policy as outlined by the instructor. Students participating in official University sponsored activities, where the student is representing the University, will meet with instructor related to absences due to university-sponsored activities. When such

absences occur, students need to provide documentation of the event(s), and work with the professor to develop a written plan for completion of missed assignments or other course requirements with as much advance notice as possible.

Communication: All university business must be conducted ***using a university assigned email account***. The use of a personal, or non-university assigned email account to conduct university business is strictly prohibited. [See full policy here](#).

Class Cancellation Policy: In the rare case that a class must be cancelled, faculty will post an announcement on the LMS that will be emailed to students. Faculty will attempt to communicate class cancellations with as much advance notice as possible. Students should be sure that their LMS email is current and valid to ensure emails are received. If the University closes due to inclement weather or other emergency situations, there will be an announcement posted on Canvas (LMS) and if possible on the local news channels (TV, apps, news websites). Communications related to University closures will also be sent to the student's cell phone number on record through the automatic University emergency text messaging system. Students should notify the University if they change their cell phone number to ensure they will receive these important emergency communications.

Academic Integrity: The University of Cincinnati is committed to academic integrity. A formal articulation of this commitment is articulated in the [Student Code of Conduct](#). The University Rules, including the Student Code of Conduct, and other documented policies of the department, college, and university related to academic integrity will be enforced. Any [violation of these regulations](#), including acts of plagiarism or cheating, will be dealt with on an individual basis according to the severity of the misconduct. At the least, you should anticipate a zero on an assignment, at the fullest extent, your violation will be reported to the university and the incident will be documented on your permanent record.

Accessibility and Special Needs: If you have a qualifying disability, UC's [Accessibility Resources](#) and your college's Access Coordinator are here to ensure you receive reasonable accommodations. Please do this as soon as possible so AR can work with me to ensure you have full and equal access to our learning environment.

I encourage you to contact [Accessibility Resources](#) to schedule a confidential meeting to discuss services and accommodations, or to begin by [registering for accommodations](#). UC uses an online system called [UC Accommodate](#), where you'll be able to register and manage your accommodations (use your UC username and password for access). Below are a few helpful UC Accommodate resources:

- [UC Accommodate Student Portal Overview](#)
- [How to Schedule an Appointment in UC Accommodate](#)
- [How to Submit a Semester Request in UC Accommodate](#)

Counseling Services: As a UC college student, there may be times when personal stressors interfere with your academic performance and/or negatively impact your daily life. If you or someone you know is experiencing challenges such as strained relationships, increased anxiety, feeling down, difficulty concentrating, lack of motivation, etc., you are not alone; there are people here that can help.

- **To schedule:** please contact Counseling and Psychological Services (CAPS) at **513-556-0648** to schedule a virtual or in-person appointment with our CAPS CECH embedded counselor, Madeleine Muller, located in the CECH Student Services Center in Suite 360 of the Teachers-Dyer Complex.
- Have questions before committing to an appointment? Check out Let's Talk, a 15-20-minute conversation to ask questions about counseling and get to know our [CECH counselor](#).
- **After-hours crisis line: 513-556-0648**
- **Self-Help:** to access Therapy Assistance Online (TAO), a **free** [self-help resource](#) for students.

Religious Accommodation: Ohio law and the University's Student Religious Accommodations for Courses Policy 1.3.7 permits a student, upon request, to be absent for reasons of faith or religious or spiritual belief system or participate in organized activities conducted under the auspices of a religious denomination, church, or other religious or spiritual organization and/or to receive alternative accommodations with regard to examinations and other course requirements due to an absence permitted for the above-described reasons. Not later than fourteen days after the first day of instruction in the course, a student should provide the instructor with written notice of the specific dates for which the student requests alternative accommodations. For additional information about this policy, please contact the Executive Director of the Office of Equal Opportunity and Access at (513) 556-5503 or oeohelp@UCMAIL.UC.EDU

Title IX: The University of Cincinnati is committed to providing a living, learning, and working environment that is free from discrimination, harassment, and retaliation. Title IX of the Education Amendments of 1972 is a federal law that prohibits discrimination on the basis of sex in education programs or activities that receive federal financial assistance. Consistent with federal, state, and local laws and regulations, the University prohibits discrimination and harassment on the basis of sex, in its academic, admissions, education and employment programs and activities. Students who wish to know more about their rights and resources on campus, they can consult the [Title IX website](#) or contact the Title IX office directly at 513-556-3349.

NOTE: The instructor reserves the right to update this syllabus as class needs arise. Be assured that they will communicate to you any changes to the schedule, syllabus or policies quickly and efficiently.